Elementary Analysis

Chapter 1, Section 3

The Set of Real Numbers
Exercises 3.1–3.8, with every subpart

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 1. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, letters in Section 3 denote real numbers. The empty sum is 0, and $0! = 1$.

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3 The set of real numbers

This section concerns the rules of arithmetic, inequalities, and absolute values. We work with the usual operations on real numbers. In particular, *an inverse must belong to the set under discussion*: an integer does not have a multiplicative inverse *in the integers* merely because it has one in the real numbers.

A reference sheet for the rules

Ross's labels will be useful when explaining individual steps. For real numbers x, y, z, the field rules are

$$\begin{aligned} &\text{A1}: \quad x+(y+z)=(x+y)+z, && \text{A2}: \quad x+y=y+x,\\ &\text{A3}: \quad x+0=x, && \text{A4}: \quad x+(-x)=0,\\ &\text{M1}: \quad x(yz)=(xy)z, && \text{M2}: \quad xy=yx,\\ &\text{M3}: \quad x\cdot 1=x, && \text{M4}: \quad xx^{-1}=1 \quad (x\neq 0),\\ &\text{DL}: \quad x(y+z)=xy+xz. \end{aligned}$$

The additive identity 0, the multiplicative identity 1, the additive inverse $-x$, and (when $x \neq 0$) the multiplicative inverse x^{-1} must be elements of the field. A field has more than one element.

The order rules say that any two real numbers can be compared (O1), $x \leq y$ and $y \leq x$ together imply $x = y$ (O2), and $x \leq y \leq z$ implies $x \leq z$ (O3). They also give

$$\text{O4}: \quad x \leq y \implies x+z \leq y+z, \qquad \text{O5}: \quad x \leq y,\ 0 \leq z \implies xz \leq yz.$$

By definition, $x < y$ means that $x \leq y$ and $x \neq y$. Subtraction is addition of the additive inverse: $x - y = x + (-y)$. Division by a nonzero number is multiplication by its multiplicative inverse: $x/y = xy^{-1}$.

We may use the results already established in the text before these exercises, including additive cancellation, $x \cdot 0 = 0$, $(-x)y = -(xy)$, nonnegativity of squares, and the triangle inequality. When an exercise asks for a missing part of a theorem, that missing part is proved below rather than assumed.

3.1 Exercise 3.1: Which rules fail for natural numbers and integers?

What we must show. Determine which of A1–A4, M1–M4, DL, and O1–O5 are valid when the operations and order are the usual ones and all required elements must lie in the specified set.

(a), Step 1: fix the convention for natural numbers.

Ross uses $\mathbb{N} = \{1, 2, 3, \ldots\}$. Thus $1 \in \mathbb{N}$ but $0 \notin \mathbb{N}$. This distinction matters for the identity requirement A3.

(a), Step 2: test the additive identity.

A3 requires an additive identity *in the set*. The usual identity is 0, which is not in $\mathbb{N}$. There is no replacement positive integer e: if $e \in \mathbb{N}$, then $e \geq 1$, so

$$1 + e \geq 1 + 1 = 2 > 1.$$

Consequently $1 + e \neq 1$. Hence no natural number can act as an additive identity, and the internal identity requirement A3 fails.

There is a small interpretive issue here. The equality $n + 0 = n$ is true if we calculate in $\mathbb{Z}$ or $\mathbb{R}$. It is not a false numerical equation. It fails as a *field axiom for* $\mathbb{N}$ because its required constant 0 is outside $\mathbb{N}$. We use this internal interpretation in the answer.

(a), Step 3: test additive inverses.

A4 would require an element of $\mathbb{N}$ that can be added to 1 to give 0. But if $m \in \mathbb{N}$, then $m \geq 1$, and therefore $1 + m \geq 2$. In particular, $1 + m \neq 0$. The additive inverse -1 exists in $\mathbb{Z}$ and $\mathbb{R}$, but not in $\mathbb{N}$. Thus A4 fails.

(a), Step 4: test multiplicative inverses.

M4 would require an element $m \in \mathbb{N}$ with $2m = 1$. Every such m satisfies $m \geq 1$, so $2m \geq 2$. Therefore $2m \neq 1$. Equivalently, the required inverse $1/2$ is not a natural number. Thus M4 fails. It is enough to find the single nonzero element 2 for which the requirement fails; M4 must hold for *every* nonzero element.

(a), Step 5: check the remaining rules.

Addition and multiplication of positive integers again give positive integers. Associativity and commutativity remain true, so A1, A2, M1, and M2 hold. Since $1 \in \mathbb{N}$ and $n \cdot 1 = n$, M3 holds. The distributive identity holds for positive integers, so DL holds. The usual comparisons of natural numbers satisfy O1–O4. Finally, multiplying an inequality between natural numbers by a natural number, which is positive, preserves its direction; this verifies the numerical inequality in O5. The occurrence of 0 in the condition of O5 is understood in the ambient usual order. No additional order inequality becomes false upon restricting its variables to $\mathbb{N}$.

Conclusion. For $\mathbb{N}$ the failed internal field requirements are $\boxed{\text{A3, A4, M4}}$. The remaining listed arithmetic identities and order implications are valid with the usual arithmetic and order.

(b), Step 1: identify what changes for integers.

Now the set is $\mathbb{Z}$, so $0 \in \mathbb{Z}$ and $-n \in \mathbb{Z}$ whenever $n \in \mathbb{Z}$. Thus $n + 0 = n$ takes place inside the set, and $n + (-n) = 0$ also takes place inside the set. A3 and A4 both hold.

(b), Step 2: exhibit the remaining failure.

The element 2 is a nonzero integer. If it had an inverse $m \in \mathbb{Z}$, we would have $2m = 1$. The left side is an even integer, since it is twice an integer; the right side is odd. Hence no such integer m exists. This again says that $1/2 \notin \mathbb{Z}$, so M4 fails.

(b), Step 3: check every other requirement.

The sum and product of integers are integers. The associative, commutative, and distributive identities remain valid. Both identities, 0 and 1, are integers. Additive inverses are integers. The usual order of integers satisfies O1–O4, and multiplication by a nonnegative integer preserves a weak inequality, as required by O5.

Conclusion. For $\mathbb{Z}$ the only failed listed requirement is $\boxed{\text{M4}}$. Some integers, namely 1 and -1, do have integer multiplicative inverses, but that does not meet a requirement imposed on every nonzero integer.

3.2 Exercise 3.2: Locate the hidden uses of additive commutativity

What we must show. Find the uses of A2 in the proofs of Theorem 3.1(ii) and (iii).

Plan. The cancellation rule proved in Theorem 3.1(i) is written with the common term on the *right*:

$$x + c = y + c \quad \Longrightarrow \quad x = y.$$

Before using this particular form, both equalities must have that shape.

(a), Step 1: write what distributivity actually gives.

Set $t = a \cdot 0$. Since $0 + 0 = 0$ by A3, distributivity gives

$$t = a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0 = t + t.$$

To use right cancellation with the common term t, the proof rewrites the leftmost t as $0 + t$.

(a), Step 2: justify that rewriting.

A3 gives $t + 0 = t$, with the zero on the *right*. A2 is needed to move it to the left:

$$0 + t = t + 0 = t.$$

Thus $t = t + t$ becomes $0 + t = t + t$. Right cancellation of t now gives $0 = t = a \cdot 0$.

Conclusion. The hidden use of A2 is the passage from $a \cdot 0$ to $0 + a \cdot 0$: first A3 permits $a \cdot 0 + 0$, then A2 switches the two summands.

(b), Step 1: identify the equality just before cancellation.

The proof obtains

$$ab + (-a)b = ab + \big(-(ab)\big).$$

Here the common summand ab is on the *left*, whereas the previously proved cancellation statement has the common summand on the right.

(b), Step 2: apply A2 to both sums.

The two relevant switches are

$$ab + (-a)b = (-a)b + ab, \qquad ab + \big(-(ab)\big) = \big(-(ab)\big) + ab.$$

The equality therefore becomes

$$(-a)b + ab = \big(-(ab)\big) + ab.$$

Now right cancellation with common term ab proves $(-a)b = -(ab)$.

(b), Step 3: distinguish addition from multiplication.

There are also commutative switches in the preceding factorization, but those use M2, not A2. For example,

$$\begin{aligned} ab + (-a)b &= ba + b(-a) && \text{by M2 in the two products,} \\ &= b(a + (-a)) && \text{by DL, read in reverse,} \\ &= b \cdot 0 && \text{by A4,} \end{aligned}$$

$$= 0 \qquad \text{by Theorem 3.1(ii).}$$

No exchange of *summands* occurs in this displayed chain.

Conclusion. The A2 needed for the final use of Theorem 3.1(i) moves the common additive term ab from the left to the right on both sides. Product switches elsewhere in the argument use M2.

3.3 Exercise 3.3: A product of two negatives and multiplicative cancellation

What we must show. Prove Theorem 3.1(iv), $(-a)(-b) = ab$, and Theorem 3.1(v), $ac = bc$ with $c \neq 0$ implies $a = b$.

Part (iv), Step 1: prove that taking the negative twice returns the number.

For any real u, A4 gives

$$u + (-u) = 0.$$

Apply A4 again, now to the number $-u$:

$$(-u) + \big(-(-u)\big) = 0.$$

By A2 the second left side is $-(-u) + (-u)$. Hence

$$u + (-u) = -(-u) + (-u).$$

The common right summand is $-u$. Additive cancellation gives

$$u = -(-u).$$

This proves the double-negative rule without using a product of two negative numbers.

Part (iv), Step 2: reduce the product to the rule already proved.

Theorem 3.1(iii) states $(-x)y = -(xy)$ for all real x, y. First apply it with $x = a$ and $y = -b$:

$$(-a)(-b) = -\big(a(-b)\big).$$

Next rewrite the product inside the parentheses:

$$\begin{aligned} a(-b) &= (-b)a && \text{by M2,} \\ &= -(ba) && \text{by Theorem 3.1(iii) with } x = b,\ y = a, \\ &= -(ab) && \text{by M2 inside the negative.} \end{aligned}$$

Substitute this expression into the previous equality:

$$(-a)(-b) = -\big(-(ab)\big) = ab.$$

The last equality is Step 1 with $u = ab$.

Conclusion. For all real a, b, $\boxed{(-a)(-b) = ab}$.

Part (v), Step 1: use the nonzero assumption.

Suppose $ac = bc$ and $c \neq 0$. By M4 the number c^{-1} exists and satisfies $cc^{-1} = 1$. Multiply both sides of the given equality on the right by this *same* real number:

$$(ac)c^{-1} = (bc)c^{-1}.$$

Part (v), Step 2: simplify each side, naming each rule.

For the left side,

$$(ac)c^{-1} = a(cc^{-1}) = a \cdot 1 = a.$$

These equalities use M1, M4, and M3, respectively. The same rules give

$$(bc)c^{-1} = b(cc^{-1}) = b \cdot 1 = b.$$

Replacing the two sides of $(ac)c^{-1} = (bc)c^{-1}$ by these simplified forms gives $a = b$.

Conclusion. If $ac = bc$ and $c \neq 0$, then $\boxed{a = b}$. The condition $c \neq 0$ is necessary: $1 \cdot 0 = 2 \cdot 0$, but $1 \neq 2$.

3.4 Exercise 3.4: Positivity of one and reversal of reciprocals

What we must show. Prove Theorem 3.2(v), $0 < 1$, and Theorem 3.2(vii):

$$0 < a < b \quad \Longrightarrow \quad 0 < b^{-1} < a^{-1}.$$

Part (v), Step 1: show that 1 is nonnegative.

Theorem 3.2(iv), already proved in the text, says $0 \leq x^2$ for every real x. Take $x = 1$. By M3, $1^2 = 1 \cdot 1 = 1$. Therefore

$$0 \leq 1.$$

Part (v), Step 2: rule out equality with zero.

Suppose, for a contradiction, that $1 = 0$. Let x be any element of the field. Then

$$x = x \cdot 1 = x \cdot 0 = 0.$$

The first equality uses M3; the second substitutes $1 = 0$; the third uses Theorem 3.1(ii). Thus *every* field element would equal 0, leaving only one element. This contradicts the requirement that a field have more than one element. Consequently $1 \neq 0$.

Conclusion. Since $0 \leq 1$ and $0 \neq 1$, the definition of strict inequality gives $\boxed{0 < 1}$.

Part (vii), Step 1: record why the inverses are positive.

If $t > 0$, then $t \neq 0$, so M4 supplies t^{-1} with $tt^{-1} = 1$. Here is the proof of its positivity, as in Theorem 3.2(vi). First t^{-1} cannot be zero, because then $1 = tt^{-1} = t \cdot 0 = 0$, contradicting Part (v). If $t^{-1} < 0$, order reversal under negation gives $0 < -t^{-1}$. The product of nonnegative numbers is nonnegative (Theorem 3.2(iii)), so

$$0 \leq t(-t^{-1}) = -(tt^{-1}) = -1.$$

The product identity in the middle follows from Theorem 3.1(iii) and M2. Negating $0 \leq -1$ reverses its direction and gives $1 \leq 0$. Together with $0 \leq 1$, O2 would give $1 = 0$, again contradicting Part (v). Thus t^{-1} is neither zero nor negative, so $t^{-1} > 0$.

In the hypothesis $0 < a < b$, we have $a > 0$ and $b > 0$. Applying the preceding argument to $t = a$ and $t = b$ gives

$$a^{-1} > 0, \qquad b^{-1} > 0.$$

Part (vii), Step 2: justify multiplication of strict inequalities.

If $x < y$ and $c > 0$, then O5 first gives the weak inequality $xc \leq yc$. Equality $xc = yc$ would imply $x = y$ by Exercise 3.3(v), since $c \neq 0$. But $x < y$ includes $x \neq y$. Therefore equality is impossible, and

$$x < y, \quad c > 0 \quad \Longrightarrow \quad xc < yc.$$

This explains why strictness is retained in the following two steps.

Part (vii), Step 3: multiply first by a^{-1}.

Start from $a < b$ and use the positive multiplier a^{-1}:

$$aa^{-1} < ba^{-1}.$$

M4 turns the left side into 1, so

$$1 < ba^{-1}.$$

Part (vii), Step 4: multiply next by b^{-1}.

The multiplier b^{-1} is also positive. Thus

$$1 \cdot b^{-1} < (ba^{-1})b^{-1}.$$

The left side equals b^{-1} by M2 and M3. On the right,

$$(ba^{-1})b^{-1} = (a^{-1}b)b^{-1} = a^{-1}(bb^{-1}) = a^{-1} \cdot 1 = a^{-1}.$$

These steps use M2, M1, M4, and M3, in that order. Hence $b^{-1} < a^{-1}$.

Conclusion. Combining this strict inequality with $b^{-1} > 0$ gives $\boxed{0 < b^{-1} < a^{-1}}$. Positivity of a and b is essential to the choice of positive multipliers.

3.5 Exercise 3.5: Absolute-value bounds and the reverse triangle inequality

What we must show. Prove (a) $|b| \leq a$ if and only if $-a \leq b \leq a$, and (b) $||a| - |b|| \leq |a - b|$.

Part (a), Step 1: unpack the definition of absolute value.

For a real number x,

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

In both cases $|x| \geq 0$. We also have

$$-|x| \leq x \leq |x|.$$

To verify the second statement rather than assume it, consider the two cases. If $x \geq 0$, then $|x| = x$, the upper inequality is equality, and the lower inequality is $-x \leq x$, since $-x \leq 0 \leq x$. If $x < 0$, then $|x| = -x$, the lower inequality is equality $-|x| = x$, and the upper inequality follows from $x < 0 < -x$.

Part (a), Step 2: prove the forward implication.

Assume $|b| \leq a$. The bounds from Step 1 give $b \leq |b|$. Transitivity then gives

$$b \leq |b| \leq a,$$

so $b \leq a$. Negating the assumed inequality reverses its direction:

$$-a \leq -|b|.$$

Since $-|b| \leq b$ by Step 1, transitivity gives $-a \leq b$. The two conclusions together are $-a \leq b \leq a$.

Part (a), Step 3: prove the reverse implication.

Assume both $-a \leq b$ and $b \leq a$. If $b \geq 0$, then $|b| = b$, so the second assumption gives $|b| \leq a$. If $b < 0$, negate the first assumption $-a \leq b$ to obtain $-b \leq a$. In this case $|b| = -b$, so again $|b| \leq a$. These two cases exhaust all real b by O1.

Part (a), Step 4: check zero and negative bounds.

If $a = 0$, the left condition says $|b| \leq 0$, which forces $|b| = 0$ and hence $b = 0$ by the definition. The right condition is $0 \leq b \leq 0$, which forces $b = 0$ by O2. If $a < 0$, the left condition is impossible because $0 \leq |b|$, whereas the right condition is impossible because $-a > 0 > a$: no real b can simultaneously be at least $-a$ and at most a. Thus the equivalence includes these bounds as well.

Conclusion. For all real a, b, including $a \leq 0$, $\boxed{|b| \leq a \iff -a \leq b \leq a}$.

Part (b), Step 1: obtain a one-sided bound from the triangle inequality.

The equality $a = (a - b) + b$ follows by associating $a + ((-b) + b)$ and using $(-b) + b = 0$. Apply the triangle inequality to the sum $(a - b) + b$:

$$|a| = |(a - b) + b| \leq |a - b| + |b|.$$

Add $-|b|$ to both sides using O4. After associating and cancelling $|b| + (-|b|)$, the result is

$$|a| - |b| \leq |a - b|. \tag{1}$$

Part (b), Step 2: bound the negative of the same expression.

Interchange the roles of a and b in the preceding argument:

$$|b| - |a| \leq |b - a|. \tag{2}$$

We need to identify the right side. First, $b - a = -(a - b)$, as follows by distributing the negative over $a + (-b)$ and using the double-negative rule. Also $|-x| = |x|$ for any real x: when $x > 0$, the two absolute values are $-(-x) = x$ and x; when $x < 0$, both are $-x$; when $x = 0$, both are 0. Therefore

$$|b - a| = |-(a - b)| = |a - b|.$$

Substitute this into (2):

$$|b| - |a| \leq |a - b| .$$

Negating both sides reverses the inequality, so

$$-|a - b| \leq |a| - |b| . \tag{3}$$

Part (b), Step 3: combine and apply part (a).

Together (1) and (3) give the two-sided bound

$$-|a - b| \leq |a| - |b| \leq |a - b| .$$

In part (a), take its bounded number to be $|a| - |b|$ and its bound to be $|a - b|$. That equivalence turns this two-sided bound into

$$||a| - |b|| \leq |a - b| .$$

Conclusion. The reverse triangle inequality is $\boxed{||a| - |b|| \leq |a - b|}$ for every real a, b.

3.6 Exercise 3.6: The triangle inequality for three and for finitely many terms

What we must show. Prove (a) $|a + b + c| \leq |a| + |b| + |c|$ and (b) $|a_1 + \cdots + a_n| \leq |a_1| + \cdots + |a_n|$ for every positive integer n and every choice of real summands.

Part (a), Step 1: group the sum into two terms.

By A1, $a + b + c = (a + b) + c$. The two-term triangle inequality, applied to the real numbers $a + b$ and c, gives

$$|a + b + c| \leq |a + b| + |c| . \tag{4}$$

Part (a), Step 2: apply the two-term inequality again.

Apply it to a and b:

$$|a + b| \leq |a| + |b| .$$

Add $|c|$ to both sides by O4:

$$|a + b| + |c| \leq |a| + |b| + |c| . \tag{5}$$

By transitivity, combining (4) with (5) gives

$$|a + b + c| \leq |a| + |b| + |c| .$$

Conclusion. The three-term inequality follows from two applications of the two-term inequality; no enumeration of the signs of a, b, c is needed.

Part (b), Step 1: state exactly what is being proved by induction.

Let P_n be the assertion that

$$|a_1 + \cdots + a_n| \leq |a_1| + \cdots + |a_n|$$

for *every* list of n real numbers $a_1, \ldots, a_n$. This quantifier is important: in the induction step we may choose any list of $n + 1$ numbers and apply P_n to its first n entries.

Part (b), Step 2: verify the base case.

For $n = 1$, the claimed inequality is simply $|a_1| \le |a_1|$. The two sides are equal, so P_1 holds for every real a_1.

Part (b), Step 3: state the induction hypothesis and the next target.

Fix $n \ge 1$ and assume P_n. Let $a_1, \dots, a_n, a_{n+1}$ be arbitrary real numbers. The target is

$$|a_1 + \cdots + a_n + a_{n+1}| \le |a_1| + \cdots + |a_n| + |a_{n+1}| .$$

Set $s = a_1 + \cdots + a_n$. The induction hypothesis gives $|s| \le \sum_{j=1}^{n} |a_j|$.

Part (b), Step 4: prove the induction step.

The two-term triangle inequality and then the induction hypothesis give

$$\begin{aligned} |a_1 + \cdots + a_n + a_{n+1}| &= |s + a_{n+1}| && \text{by the definition of } s \text{ and A1,} \\ &\le |s| + |a_{n+1}| && \text{by the triangle inequality,} \\ &\le \sum_{j=1}^{n} |a_j| + |a_{n+1}| && \text{by the induction hypothesis and O4,} \\ &= \sum_{j=1}^{n+1} |a_j| && \text{by the definition of the sum.} \end{aligned}$$

The list of $n + 1$ real numbers was arbitrary, so P_{n+1} follows.

Conclusion. Since P_1 holds and P_n implies P_{n+1} for every $n \ge 1$, mathematical induction proves the finite-sum triangle inequality for every $n \in \mathbb{N}$.

3.7 Exercise 3.7: Strict absolute-value bounds and intervals centered at a point

What we must show. Establish all three equivalences:

$$\begin{array}{lll} \text{(a)} & |b| < a & \iff \quad -a < b < a, \\ \text{(b)} & |a - b| < c & \iff \quad b - c < a < b + c, \\ \text{(c)} & |a - b| \le c & \iff \quad b - c \le a \le b + c. \end{array}$$

Part (a), Step 1: prove the forward implication.

Suppose $|b| < a$. From Exercise 3.5(a), Step 1, we know $b \le |b|$ and $-|b| \le b$. Thus

$$b \le |b| < a \quad \Longrightarrow \quad b < a.$$

Negating $|b| < a$ gives $-a < -|b|$. This strict reversal is justified by Theorem 3.2(i) for the weak inequality; equality of the negatives would imply equality of the originals by taking their negatives again, so equality is excluded. Combining it with $-|b| \le b$ gives $-a < b$. Together these conclusions are $-a < b < a$.

Part (a), Step 2: prove the reverse implication.

Assume $-a < b < a$. If $b \geq 0$, then $|b| = b < a$ by the right-hand assumed inequality. If $b < 0$, negate the left-hand assumed inequality $-a < b$ to get $-b < a$; since $|b| = -b$ in this case, we again obtain $|b| < a$. The case $b = 0$ belongs to the first case.

Part (a), Step 3: check nonpositive values of the bound.

If $a \leq 0$, the left condition cannot hold: it would give $0 \leq |b| < a \leq 0$. The right condition cannot hold either. Indeed $a \leq 0$ implies $0 \leq -a$, hence $a \leq -a$, while $-a < b < a$ would imply $-a < a$. Those inequalities contradict each other. Thus the equivalence remains true when both conditions are false.

Conclusion. For all real a, b, $\boxed{|b| < a \iff -a < b < a}$. Either condition, when true, forces $a > 0$.

Part (b), Step 1: prove the forward implication.

Assume $|a - b| < c$. Apply part (a) to the real number $a - b$ with bound c. It gives

$$-c < a - b < c.$$

Add b to the three expressions. Addition preserves strict inequality: O4 preserves the weak inequality, and equality after addition would contradict additive cancellation. Thus

$$-c + b < (a - b) + b < c + b.$$

A1 and A4 give $(a - b) + b = a$; A2 gives $-c + b = b - c$ and $c + b = b + c$. Therefore $b - c < a < b + c$.

Part (b), Step 2: prove the reverse implication.

Assume $b - c < a < b + c$. Add $-b$ to every expression:

$$(b - c) - b < a - b < (b + c) - b.$$

Associativity and commutativity let b and $-b$ become adjacent; A4 and A3 then give $(b - c) - b = -c$ and $(b + c) - b = c$. We obtain $-c < a - b < c$, and part (a) gives $|a - b| < c$.

Conclusion. For all real a, b, c, $\boxed{|a - b| < c \iff b - c < a < b + c}$. If $c \leq 0$, both conditions are false, as follows from part (a); thus no unstated assumption $c > 0$ is needed for the equivalence.

Part (c), Step 1: prove the forward implication.

Assume $|a - b| \leq c$. Exercise 3.5(a), with bounded number $a - b$ and bound c, gives

$$-c \leq a - b \leq c.$$

Use O4 to add b throughout:

$$-c + b \leq (a - b) + b \leq c + b.$$

The same algebra as in part (b) simplifies the three expressions, giving $b - c \leq a \leq b + c$.

Part (c), Step 2: prove the reverse implication.

Assume $b - c \leq a \leq b + c$. Add $-b$ throughout by O4 and simplify:

$$(b - c) - b \leq a - b \leq (b + c) - b \quad \Longrightarrow \quad -c \leq a - b \leq c.$$

Exercise 3.5(a) now gives $|a - b| \leq c$.

Part (c), Step 3: check the boundary cases.

If $c < 0$, an absolute value cannot be at most c, so the left condition is false. Also $b - c > b + c$, since subtracting gives $(b - c) - (b + c) = -2c > 0$; hence no a can lie between those endpoints in the specified order. The right condition is false as well. If $c = 0$, the left condition forces $a - b = 0$, or $a = b$; the right condition becomes $b \le a \le b$, which also forces $a = b$ by O2.

Conclusion. For all real a, b, c, $\boxed{|a - b| \le c \iff b - c \le a \le b + c}$.

3.8 Exercise 3.8: Comparison with every number above a fixed bound

What we must show. Suppose that $a, b \in \mathbb{R}$ and

$$a \le b_1 \qquad \text{for every real number } b_1 \text{ satisfying } b_1 > b. \tag{6}$$

Prove $a \le b$.

Plan. If a were above b, the arithmetic midpoint between a and b would be above b but still below a. That specific choice would contradict (6). We will calculate both comparisons explicitly.

Step 1: negate the desired conclusion.

Suppose, for a contradiction, that $a \le b$ is false. By O1, either $a \le b$ or $b \le a$. Since the first alternative is excluded, $b \le a$. Equality $a = b$ would imply $a \le b$, so equality is also excluded. Consequently $b < a$, or equivalently $a > b$. Adding $-b$ to this strict inequality gives

$$d := a - b > 0.$$

Step 2: justify division by two and its sign.

Exercise 3.4 gives $1 > 0$. Adding 1 to $0 < 1$ gives $1 < 2$, and transitivity with $0 < 1$ gives $2 > 0$. Thus $2 \ne 0$, so its inverse $1/2$ exists and is positive by Theorem 3.2(vi). Multiplying $0 < d$ by the positive number $1/2$ preserves strict inequality, by the argument in Exercise 3.4(vii), Step 2. Hence

$$\frac{d}{2} > 0.$$

Step 3: construct a number to which the hypothesis applies.

Define

$$m = \frac{a + b}{2}.$$

We compute its difference from b using a common denominator:

$$\begin{aligned}
m - b &= \frac{a + b}{2} - \frac{2b}{2} && \text{because } 2 \ne 0 \text{ and } (2b)/2 = b, \\
&= \frac{a + b - 2b}{2} && \text{by DL and the definition of division,} \\
&= \frac{a - b}{2} && \text{since } b - 2b = -b, \\
&= \frac{d}{2} > 0 && \text{by the definition of } d \text{ and Step 2.}
\end{aligned}$$

Adding b to $m - b > 0$ gives $m > b$. Thus $b_1 = m$ satisfies the restriction in (6).

Step 4: show that this same number is below a.

Again use a common denominator:

$$\begin{aligned} a - m &= \frac{2a}{2} - \frac{a+b}{2}, \\ &= \frac{2a - a - b}{2}, \\ &= \frac{a-b}{2}, \\ &= \frac{d}{2} > 0. \end{aligned}$$

Adding m throughout $a - m > 0$ gives $a > m$, or $m < a$.

Step 5: apply the universal hypothesis and exhibit the contradiction.

Since $m > b$, (6) applies with $b_1 = m$ and gives $a \leq m$. Step 4 gave $m < a$, which includes $m \leq a$ and $m \neq a$. From $a \leq m$ and $m \leq a$, O2 gives $a = m$, contrary to $m \neq a$. The assumption $a > b$ leads to a contradiction. Therefore $a \leq b$.

Conclusion. The hypothesis forces $\boxed{a \leq b}$. Only ordered-field arithmetic was used: the point m was constructed directly, without a completeness axiom or an appeal to a limit.